

combined_cv_final

scalars_ensemble_cv

key	value
skipped_empty_gt	0
main_kwargs_applied.substitute	False
main_kwargs_applied.normalize_final_letters	True
step2_stats.n_models	9
step2_stats.spearman	1
step2_stats.kendall_b	1
step2_stats.spearman_exact_p_onetailed	2.7557e-06
step2_stats.kendall_exact_p_onetailed	2.7557e-06
step2_stats.p_method	exact
step2_stats.n_permutations	362880
step2_stats.min_attainable_p	2.7557e-06
step2_stats.pearson	0.9953
step2_stats.pearson_p	2.3346e-08
step2_stats.mean_wer_compression	0.9027
step3_determinism.identical_outputs	10495
step3_determinism.n	10495
step3_determinism.deterministic	True
step3_determinism.main_kwargs.substitute	False
step3_determinism.main_kwargs.normalize_final_letters	True
step4_resolution_true.reference	true
step4_resolution_true.n_boot	10000
step4_resolution_true.n_samples	10500
step4_resolution_true.min_resolved_gap_wer	0.6214
step4_resolution_true.max_unresolved_gap_wer	0
step4_resolution_true.n_resolved	36
step4_resolution_true.n_pairs	36
step4_resolution_pseudo.reference	pseudo
step4_resolution_pseudo.n_boot	10000
step4_resolution_pseudo.n_samples	10495
step4_resolution_pseudo.min_resolved_gap_wer	0.6799
step4_resolution_pseudo.max_unresolved_gap_wer	0
step4_resolution_pseudo.n_resolved	36
step4_resolution_pseudo.n_pairs	36

step0_cross_wer_ensemble_cv

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	15.8	19.4	11.85	24.88	25.77	13.86	30.72	10.23
whisper_v3	15.8	0	14.77	15	25.5	26.23	17.06	31.71	13.88
whisper_v2	19.4	14.77	0	18.63	28.03	28.83	20.5	33.99	17.57
fastconformer	11.85	15	18.63	0	22.77	24.06	10.3	28.8	7.05
xlsr	24.88	25.5	28.03	22.77	0	31.68	23.49	31.1	22.29
w2vbert	25.77	26.23	28.83	24.06	31.68	0	25.33	37.2	23.32
hubert	13.86	17.06	20.5	10.3	23.49	25.33	0	29.58	9.23
wav2vec_sb	30.72	31.71	33.99	28.8	31.1	37.2	29.58	0	28.81
cohere	10.23	13.88	17.57	7.05	22.29	23.32	9.23	28.81	0

step0_err_corr_ensemble_cv

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	1	0.493	0.4572	0.4468	0.3933	0.3559	0.3527	0.3807	0.4864
whisper_v3	0.493	1	0.6827	0.4367	0.4573	0.439	0.3487	0.4105	0.4442
whisper_v2	0.4572	0.6827	1	0.3904	0.4453	0.4221	0.3313	0.4116	0.4045
fastconformer	0.4468	0.4367	0.3904	1	0.41	0.3553	0.4155	0.4007	0.5511
xlsr	0.3933	0.4573	0.4453	0.41	1	0.4547	0.4217	0.6335	0.3863
w2vbert	0.3559	0.439	0.4221	0.3553	0.4547	1	0.3421	0.4296	0.3379
hubert	0.3527	0.3487	0.3313	0.4155	0.4217	0.3421	1	0.4044	0.3997
wav2vec_sb	0.3807	0.4105	0.4116	0.4007	0.6335	0.4296	0.4044	1	0.3594
cohere	0.4864	0.4442	0.4045	0.5511	0.3863	0.3379	0.3997	0.3594	1

step0_flags_ensemble_cv

pair	cross_wer	err_corr	note
seamless_m4t fastconformer	11.85	0.4468	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t hubert	13.86	0.3527	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t cohere	10.23	0.4864	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v3 cohere	13.88	0.4442	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer hubert	10.3	0.4155	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer cohere	7.05	0.5511	unusually similar hypotheses - suspect shared lab/data/teacher
hubert cohere	9.23	0.3997	unusually similar hypotheses - suspect shared lab/data/teacher

step1_bias_ensemble_cv

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
cohere	voter	10495	0.5884	0.582	0.5636	0.5535	-0.0064	-0.01
fastconformer	voter	10495	0.7076	0.7083	0.6769	0.6751	0.0008	-0.0019
hubert	voter	10495	0.7213	0.7074	0.691	0.6738	-0.0139	-0.0172
seamless_m4t	voter	10495	0.798	0.8061	0.7631	0.7689	0.0081	0.0058
whisper_v3	voter	10495	0.8672	0.8813	0.8275	0.8438	0.0141	0.0163
whisper_v2	ranked_only(control)	10495	0.9035	0.9035	0.8642	0.8642	0	0
w2vbert	voter	10495	0.9228	0.9294	0.8801	0.8885	0.0066	0.0084
xlsr	ranked_only(control)	10495	0.928	0.928	0.8877	0.8877	0	0
wav2vec_sb	voter	10495	0.9519	0.9561	0.9144	0.9206	0.0041	0.0062

step2_ranking_ensemble_cv

model	role	n_samples	wer_true	cer_true	wer_pseudo_loo	cer_pseudo_loo	rank_true	rank_pseudo	wer_compression
cohere	voter	10496	5.5591	1.5028	5.4715	1.6081	1	1	0.0876
fastconformer	voter	10496	7.1898	2.158	6.864	2.0614	2	2	0.3258
hubert	voter	10495	7.8081	2.1291	9.0171	2.4024	3	3	-1.209
seamless_m4t	voter	10495	10.7671	3.4897	9.697	3.3403	4	4	1.0701
whisper_v3	voter	10495	14.7106	4.636	13.0205	4.1906	5	5	1.6901
whisper_v2	ranked_only	10495	18.4155	6.0072	16.4118	5.4923	6	6	2.0037
xlsr	ranked_only	10495	22.8174	6.433	21.3218	5.9627	7	7	1.4956
w2vbert	voter	10495	23.9822	6.5829	22.5853	6.2034	8	8	1.3969
wav2vec_sb	voter	10495	29.1187	9.6559	27.8555	9.2256	9	9	1.2632

step3_model_dropout_ensemble_cv

model	times_filtered_out	times_in_pool	filtered_out_rate
wav2vec_sb	39018	73467	0.5311
w2vbert	29852	73467	0.4063
whisper_v3	13295	73467	0.181
seamless_m4t	8429	73467	0.1147
hubert	6275	73467	0.0854
fastconformer	4214	73466	0.0574
cohere	2518	73466	0.0343

step3_runs_ensemble_cv

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_longest_match_validation	refsel_most_common_words
full_pool	7	10495	0.8566	0.0406	0.7521	0.2479

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_longest_match_validation	refsel_most_common_words
LOO_minus_seamless_m4t	6	10495	0.8351	0.0339	0.7783	0.2217
LOO_minus_whisper_v3	6	10495	0.8309	0.0303	0.7889	0.2111
LOO_minus_fastconformer	6	10496	0.8407	0.0364	0.7772	0.2228
LOO_minus_w2vbert	6	10495	0.7747	0.0257	0.8178	0.1822
LOO_minus_hubert	6	10495	0.8411	0.0349	0.7833	0.2167
LOO_minus_wav2vec_sb	6	10495	0.7482	0.025	0.8333	0.1667
LOO_minus_cohere	6	10496	0.8413	0.0403	0.7713	0.2287

step4_pairs_pseudo_ensemble_cv

pair	adjacent	gap_wer	preserved_frac	resolved_95
hubert < seamless_m4t	True	0.6799	0.9996	True
xlsr < w2vbert	True	1.2636	1	True
cohere < fastconformer	True	1.3925	1	True
fastconformer < hubert	True	2.1548	1	True
fastconformer < seamless_m4t	False	2.8347	1	True
seamless_m4t < whisper_v3	True	3.3235	1	True
whisper_v3 < whisper_v2	True	3.3913	1	True
cohere < hubert	False	3.5474	1	True
hubert < whisper_v3	False	4.0034	1	True
cohere < seamless_m4t	False	4.2273	1	True
whisper_v2 < xlsr	True	4.91	1	True
w2vbert < wav2vec_sb	True	5.2702	1	True
fastconformer < whisper_v3	False	6.1582	1	True
whisper_v2 < w2vbert	False	6.1735	1	True
xlsr < wav2vec_sb	False	6.5337	1	True
seamless_m4t < whisper_v2	False	6.7148	1	True
hubert < whisper_v2	False	7.3947	1	True
cohere < whisper_v3	False	7.5508	1	True
whisper_v3 < xlsr	False	8.3013	1	True
fastconformer < whisper_v2	False	9.5495	1	True
whisper_v3 < w2vbert	False	9.5648	1	True
cohere < whisper_v2	False	10.9421	1	True
whisper_v2 < wav2vec_sb	False	11.4437	1	True
seamless_m4t < xlsr	False	11.6248	1	True
hubert < xlsr	False	12.3047	1	True
seamless_m4t < w2vbert	False	12.8883	1	True
hubert < w2vbert	False	13.5682	1	True
fastconformer < xlsr	False	14.4595	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v3 < wav2vec_sb	False	14.835	1	True
fastconformer < w2vbert	False	15.7231	1	True
cohere < xlsr	False	15.852	1	True
cohere < w2vbert	False	17.1156	1	True
seamless_m4t < wav2vec_sb	False	18.1585	1	True
hubert < wav2vec_sb	False	18.8384	1	True
fastconformer < wav2vec_sb	False	20.9932	1	True
cohere < wav2vec_sb	False	22.3858	1	True

step4_pairs_true_ensemble_cv

pair	adjacent	gap_wer	preserved_frac	resolved_95
fastconformer < hubert	True	0.6214	1	True
xlsr < w2vbert	True	1.1621	1	True
cohere < fastconformer	True	1.6296	1	True
cohere < hubert	False	2.2511	1	True
hubert < seamless_m4t	True	2.9664	1	True
fastconformer < seamless_m4t	False	3.5878	1	True
whisper_v3 < whisper_v2	True	3.7042	1	True
seamless_m4t < whisper_v3	True	3.9295	1	True
whisper_v2 < xlsr	True	4.3988	1	True
w2vbert < wav2vec_sb	True	5.1348	1	True
cohere < seamless_m4t	False	5.2174	1	True
whisper_v2 < w2vbert	False	5.561	1	True
xlsr < wav2vec_sb	False	6.2969	1	True
hubert < whisper_v3	False	6.8958	1	True
fastconformer < whisper_v3	False	7.5173	1	True
seamless_m4t < whisper_v2	False	7.6337	1	True
whisper_v3 < xlsr	False	8.103	1	True
cohere < whisper_v3	False	9.1469	1	True
whisper_v3 < w2vbert	False	9.2652	1	True
hubert < whisper_v2	False	10.6	1	True
whisper_v2 < wav2vec_sb	False	10.6958	1	True
fastconformer < whisper_v2	False	11.2215	1	True
seamless_m4t < xlsr	False	12.0325	1	True
cohere < whisper_v2	False	12.8511	1	True
seamless_m4t < w2vbert	False	13.1947	1	True
whisper_v3 < wav2vec_sb	False	14.4	1	True
hubert < xlsr	False	14.9989	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
fastconformer < xlsr	False	15.6203	1	True
hubert < w2vbert	False	16.161	1	True
fastconformer < w2vbert	False	16.7824	1	True
cohere < xlsr	False	17.2499	1	True
seamless_m4t < wav2vec_sb	False	18.3295	1	True
cohere < w2vbert	False	18.4121	1	True
hubert < wav2vec_sb	False	21.2958	1	True
fastconformer < wav2vec_sb	False	21.9172	1	True
cohere < wav2vec_sb	False	23.5469	1	True

step5_baselines_ensemble_cv

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble (proposed)	9	1	1	2.7557e-06	2.7557e-06	exact	362880	2.7557e-06	1.1713
full ensemble (biased)	9	1	1	2.7557e-06	2.7557e-06	exact	362880	2.7557e-06	1.83
single model: seamless_m4t	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	2.8048
LOO ensemble matched to n-1 (vs seamless_m4t)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.184
single model: whisper_v3	8	0.8333	0.7143	0.0077	0.0071	exact	40320	2.4802e-05	5.1675
LOO ensemble matched to n-1 (vs whisper_v3)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1065
single model: fastconformer	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.7331
LOO ensemble matched to n-1 (vs fastconformer)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.277
single model: w2vbert	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	13.2048
LOO ensemble matched to n-1 (vs w2vbert)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1431
single model: hubert	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	2.003
LOO ensemble matched to n-1 (vs hubert)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1666
single model: wav2vec_sb	8	0.9286	0.8571	0.0011	0.0009	exact	40320	2.4802e-05	17.8119
LOO ensemble matched to n-1 (vs wav2vec_sb)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1598
single model: cohere	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.7447
LOO ensemble matched to n-1 (vs cohere)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.3068

step6_contribution_ensemble_cv

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
seamless_m4t	0.2499	0	True	True	0	0
whisper_v3	-0.1672	0	False	True	0	0
fastconformer	0.5072	0	True	True	0	0
w2vbert	-0.0864	0	False	True	0	0
hubert	0.6989	0	True	True	0	0

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
wav2vec_sb	-0.1221	0	False	True	0	0
cohere	0.9036	0	True	True	0	0

step6_pools_ensemble_cv

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+fastconformer+hubert+cohere	4	4.4507	1.2852	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+cohere	5	4.4883	1.2902	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+hubert+wav2vec_sb+cohere	5	4.5315	1.3115	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+cohere	5	4.586	1.2818	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	4.6424	1.3267	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+hubert+cohere	3	4.6461	1.2982								
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+cohere	6	4.6875	1.2841	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	6	4.7232	1.3183	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+cohere	4	4.8021	1.3613	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	7	4.8096	1.3674	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+cohere	4	4.8434	1.4416	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+hubert+wav2vec_sb+cohere	4	4.8603	1.3937	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+hubert+cohere	3	4.8776	1.387								
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	4.911	1.4564	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	4.9392	1.4758	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	4.9467	1.4309	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+hubert+wav2vec_sb+cohere	4	5.0481	1.4203	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	5.0595	1.4823	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+w2vbert+hubert+cohere	4	5.0612	1.4168	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+fastconformer+hubert	3	5.1402	1.4701								
seamless_m4t+fastconformer+cohere	3	5.1969	1.4737								
seamless_m4t+whisper_v3+w2vbert+hubert+cohere	5	5.2209	1.4621	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+hubert+cohere	4	5.2399	1.4946	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9664	hubert seamless_m4t
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	5.2566	1.4762	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	5.2624	1.4671	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	5.3168	1.4731	1	2.7557e-06	1	2.7557e-06	0	0	0	

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_t rue	ranking_spear man	ranking_exact _p	ranking_kendall _b	ranking_kendall_exact _p	n_pairs_wr ong	n_pairs_wrong_resolva ble	worst_wrong_gap _wer	worst_wrong_pair
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	5.3243	1.4971	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+cohere	4	5.3262	1.4758	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	5.4145	1.5409	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert	4	5.4464	1.558	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+hubert+cohere	3	5.4577	1.5767								
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	5.499	1.6086	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert	4	5.5084	1.6128	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	6	5.5085	1.5451	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb+cohere	5	5.5329	1.5736	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+cohere	5	5.5348	1.5489	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+cohere	4	5.5423	1.5859	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert	5	5.5873	1.6272	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb	5	5.5873	1.6588	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb	5	5.6004	1.6383	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+hubert+wav2vec_sb+cohere	4	5.6587	1.6086	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.8948	hubert seamless_m4t
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	6	5.7132	1.6664	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+w2vbert+hubert+cohere	4	5.7132	1.6017	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.8948	hubert seamless_m4t
whisper_v3+fastconformer+cohere	3	5.7507	1.6562								
fastconformer+w2vbert+cohere	3	5.8523	1.6821								
whisper_v3+fastconformer+wav2vec_sb+cohere	4	5.8636	1.6748	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert	3	5.8766	1.7254								
w2vbert+hubert+cohere	3	5.8992	1.7322								
whisper_v3+fastconformer+w2vbert+cohere	4	5.965	1.6843	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5555	fastconformer hubert
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	5.9687	1.7231	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.9571	hubert seamless_m4t
fastconformer+w2vbert+wav2vec_sb+cohere	4	5.9951	1.7833	0.8667	0.0023	0.7778	0.0012	4	4	5.208	cohere seamless_m4t
fastconformer+wav2vec_sb+cohere	3	6.1172	1.8259								
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	6.1172	1.7795	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+wav2vec_sb	4	6.149	1.8506	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
w2vbert+hubert+wav2vec_sb+cohere	4	6.1584	1.8205	0.9167	0.0007	0.8333	0.0004	3	3	5.2097	cohere seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert	4	6.1866	1.8369	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
hubert+wav2vec_sb+cohere	3	6.4045	2.0164								
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	6.412	1.9339	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
fastconformer+w2vbert+hubert	3	6.4346	1.9175								

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	6.63	1.8734	0.9333	0.0004	0.8333	0.0004	3	3	3.9434	seamless_m4t whisper_v3
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	6.6657	1.8742	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
fastconformer+w2vbert+hubert+wav2vec_sb	4	6.7032	2.0503	0.95	0.0002	0.8889	0.0001	2	2	3.5789	fastconformer seamless_m4t
seamless_m4t+whisper_v3+w2vbert+cohere	4	6.7043	1.855	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6296	cohere fastconformer
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	6.7081	1.9386	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6296	cohere fastconformer
seamless_m4t+w2vbert+cohere	3	6.71	1.9379								
seamless_m4t+whisper_v3+cohere	3	6.7569	1.9805								
seamless_m4t+wav2vec_sb+cohere	3	6.817	2.0557								
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb	5	6.9888	2.037	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+hubert	3	7.0085	2.101								
seamless_m4t+whisper_v3+w2vbert+hubert	4	7.0113	1.9647	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.8948	hubert seamless_m4t
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	7.017	2.0811	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+hubert+wav2vec_sb	3	7.0207	2.228								
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	7.1166	2.0636	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.6219	fastconformer hubert
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	7.1259	2.0473	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5591	fastconformer hubert
seamless_m4t+whisper_v3+fastconformer	3	7.1568	2.1261								
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	7.1767	2.0183	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5591	fastconformer hubert
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	7.1973	2.115	0.9667	8.2672e-05	0.8889	0.0001	2	2	3.9434	seamless_m4t whisper_v3
seamless_m4t+fastconformer+w2vbert	3	7.2142	2.0979								
seamless_m4t+w2vbert+hubert	3	7.3552	2.1641								
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	7.3608	2.2006	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	3.9434	seamless_m4t whisper_v3
seamless_m4t+fastconformer+wav2vec_sb	3	7.4472	2.2812								
seamless_m4t+hubert+wav2vec_sb	3	7.7703	2.4681								
whisper_v3+w2vbert+wav2vec_sb+cohere	4	8.2024	2.403	0.9	0.001	0.8333	0.0004	3	3	5.2097	cohere seamless_m4t
whisper_v3+wav2vec_sb+cohere	3	8.5686	2.6572								
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	8.7003	2.5833	0.95	0.0002	0.8889	0.0001	2	2	3.5789	fastconformer seamless_m4t
whisper_v3+w2vbert+cohere	3	8.7376	2.5797								
whisper_v3+w2vbert+hubert+wav2vec_sb	4	8.9427	2.6963	0.9667	8.2672e-05	0.8889	0.0001	2	2	3.7067	whisper_v3 whisper_v2
whisper_v3+fastconformer+wav2vec_sb	3	9.1437	2.8478								
whisper_v3+fastconformer+w2vbert	3	9.247	2.7214								
whisper_v3+w2vbert+hubert	3	9.3391	2.7602								
whisper_v3+hubert+wav2vec_sb	3	9.4518	3.0091								
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	9.5307	2.7755	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	3.7067	whisper_v3 whisper_v2
seamless_m4t+whisper_v3+w2vbert	3	10.1138	3.0177								
seamless_m4t+whisper_v3+wav2vec_sb	3	10.1288	3.1773								
w2vbert+wav2vec_sb+cohere	3	10.6072	3.3572								
fastconformer+w2vbert+wav2vec_sb	3	11.0618	3.5151								

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
w2vbert+hubert+wav2vec_sb	3	11.3418	3.6772								
seamless_m4t+w2vbert+wav2vec_sb	3	12.0764	3.8016								
whisper_v3+w2vbert+wav2vec_sb	3	13.9682	4.4107								

step6_wrong_pairs_ensemble_cv

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7688	13.0841	11.1988	5.2097	1.8854	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7688	11.3152	10.9194	5.2097	0.3957	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7671	10.5566	10.4267	5.208	0.1299	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	14.3422	13.5622	3.9434	0.78	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	14.981	14.1028	3.9434	0.8782	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	13.8247	13.2455	3.9434	0.5792	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7122	18.4189	16.2232	16.1696	3.7067	0.0536	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7122	18.4189	16.9624	16.8765	3.7067	0.0859	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7688	13.9701	11.7064	3.5789	2.2637	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7688	12.1235	11.4876	3.5789	0.636	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7671	11.1419	10.4267	3.5773	0.7152	True
seamless_m4t+whisper_v3+hubert+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.8403	10.1675	10.0352	2.9664	0.1323	True
whisper_v3+fastconformer+w2vbert+hubert	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.8477	10.558	2.9571	1.2896	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.5809	10.6218	2.9571	0.9591	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	15.5799	12.2245	2.9571	3.3554	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	13.0245	11.4876	2.9571	1.5369	True
w2vbert+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	12.7742	10.9194	2.9571	1.8547	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.2581	10.6041	2.9571	0.654	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.0538	10.1752	2.9571	0.8786	True
seamless_m4t+whisper_v3+w2vbert+hubert	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	12.7377	12.5242	2.8948	0.2134	True
whisper_v3+w2vbert+hubert+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	11.5678	10.0767	2.8948	1.4911	True
whisper_v3+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	11.2874	10.1382	2.8948	1.1492	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8117	11.3085	9.8495	2.2526	1.459	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8117	13.0841	11.0538	2.2526	2.0303	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8081	10.5566	9.2385	2.249	1.3182	True
seamless_m4t+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.3153	6.8991	1.6307	0.4162	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	11.3085	7.81	1.6307	3.4985	True
seamless_m4t+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.8138	6.8553	1.6307	0.9585	True
whisper_v3+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	9.0556	7.2947	1.6307	1.761	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	13.0841	8.956	1.6307	4.1281	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	9.2973	7.1862	1.6307	2.1111	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	11.3152	7.6987	1.6307	3.6165	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	8.693	7.793	1.6307	0.9	True
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.3296	6.9078	1.6307	0.4219	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	8.6904	7.3303	1.6307	1.3601	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	9.2944	7.9184	1.6296	1.376	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	9.2058	7.8134	1.6296	1.3924	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	12.3979	10.1523	0.6219	2.2456	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	13.9701	11.2581	0.6219	2.712	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	10.1032	10.0874	0.6219	0.0158	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer hubert	fastconformer	hubert	7.1898	7.8081	11.1419	9.2385	0.6182	1.9034	True
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	10.6988	10.2778	0.5591	0.421	True
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	10.5395	10.175	0.5591	0.3645	True
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	9.4957	9.3612	0.5555	0.1345	True

step7_floor_ensemble_cv

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
cohere	0.8011	0.5538	0.323	2088	0.5708
hubert	0.7598	0.6738	0.5908	2521	0.7006
fastconformer	0.757	0.6752	0.4801	2551	0.6871
seamless_m4t	0.6598	0.7689	0.594	3570	0.7858
whisper_v3	0.5688	0.8438	0.7158	4525	0.8539
whisper_v2	0.4936	0.8642	0.7573	5315	0.8738
xlsr	0.4227	0.8877	0.8047	6059	0.9068
w2vbert	0.3718	0.8885	0.8204	6593	0.9062
wav2vec_sb	0.317	0.9206	0.8733	7168	0.9366